

PURBANCHAL UNIVERSITY

2026

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

MCA102: Software Engineering (New Course)

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group A

2×12=24

Answer TWO questions.

1. Define software engineering with ethical issues. Explain waterfall and spiral software models with its applications. •
2. Why is it important to estimate various project parameters before starting to develop a software? Explain COCOMO model with reference to project cost estimation. •
3. What do you mean by architectural designs? Explain few important functions of architectural designs. What is the expected output of software architectural design? •

Group B

7×8=56

Answer SEVEN questions.

4. Explain the software requirement analysis and specification. Discuss various methods for requirement gathering. •
5. What is the process of software testing? Explain the different testing methods illustrating their importance. •
6. Explain the concept of formal technical review with reference to software quality. Why software reliability is crucial for success of a software? •
7. What is the purpose of software reuse? What are the factors influencing software reuse? •
8. Explain how cloud computing technologies is enhancing software development and implementation. List few important features of cloud technology. •
9. Explain win-win spiral model with its salient features and a neat diagram. •

Contd. ...

(2)

10. Differentiate between functional and non-functional requirement. What steps are followed during SRS preparation. •
11. Write short notes on (Any TWO): •
- (a) Agile methodology.
 - (b) Object Oriented software engineering.
 - (c) Software QoS.

2×3=6

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